Exercise Sheet 3

Generative Models and Naive Bayes

Probabilistic Machine Learning

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3.A A Posterior and Its Peak

Exercise 3.A.1: Two Analysts, One Campaign

Same data: 60 of 400 converted. Analyst A uses a flat prior $\mathrm{Beta}(1, 1)$; analyst B, from earlier campaigns, uses $\mathrm{Beta}(20, 80)$.

1. Write down both posteriors and their means.
2. How many "pseudo-observations" is B's prior worth? Why do the two analysts nearly agree?
3. The data arrived in two batches: 25 of the first 200, 35 of the second 200. Show that updating twice gives the same posterior as updating once.
4. What is A's probability that the next user converts?

Exercise 3.A.1: Two Analysts, One Campaign — Solution

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1. A: $\mathrm{Beta}(61, 341)$, mean $61/402 \approx 0.152$. B: $\mathrm{Beta}(80, 420)$, mean $80/500 = 0.160$. 95% credible intervals: A $[0.118, 0.188]$, B $[0.129, 0.193]$.
2. $a + b = 100$ pseudo-observations against 400 real ones; B's prior mean 0.2 is pulled almost all the way to the data's 0.15.
3. A: $\mathrm{Beta}(1, 1) \to \mathrm{Beta}(26, 176) \to \mathrm{Beta}(61, 341)$ — counts simply add, in any order.
4. The posterior mean, ≈ 0.152: averaging θ over the posterior.

Exercise 3.A.2: Peak, Mean and a Prior's Peak

Returning users, $\theta = P(\text{user returns})$, prior $\text{Beta}(a, b)$, h of n users return.

1. Maximise $\log p(D \mid \theta) + \log p(\theta)$ and show $\hat{\theta}_{\text{MAP}} = \frac{h+a-1}{n+a+b-2}$.
2. **0 of 4** users return. Give the MLE, the MAP under $\text{Beta}(2, 2)$ and the posterior mean under $\text{Beta}(2, 2)$. Which one would you report to a product manager, and why?
3. The posterior mean under a flat prior is $\frac{h+1}{n+2}$. Under which prior is the **MAP** equal to it?
4. After 3 of 3 with a flat prior the posterior is $\text{Beta}(4, 1)$, with density $4\theta^3$. Compute its central 95% interval.

Exercise 3.A.2: Peak, Mean and a Prior's Peak — Solution

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1. $\log p(D \mid \theta) + \log p(\theta) = (h + a - 1)\log\theta + (n - h + b - 1)\log(1 - \theta) + \text{const}$. Setting the derivative $\frac{h+a-1}{\theta} - \frac{n-h+b-1}{1-\theta}$ to zero gives the formula — the MLE of a dataset with $a - 1$ extra yeses and $b - 1$ extra noes.
2. MLE $0/4 = 0$: "nobody ever returns". MAP $\frac{0+1}{4+2} = \frac{1}{6} \approx 0.17$. Posterior $\text{Beta}(2, 6)$, mean $\frac{2}{8} = 0.25$. Report the posterior (mean and interval): four users are too few for a point, and the MLE claims certainty the data cannot give.
3. $\frac{h+a-1}{n+a+b-2} = \frac{h+1}{n+2}$ for all h, n needs $a = b = 2$: the MAP under $\text{Beta}(2, 2)$ equals the posterior mean under $\text{Beta}(1, 1)$. "Add one yes and one no" can be read either way — the Naive Bayes module of Lecture 03 calls it Laplace smoothing.
4. The CDF is θ^4, so the interval is $[0.025^{1/4}, 0.975^{1/4}] \approx [0.40, 0.99]$ — the peak at 1 lies at the edge of it.

3.B The Generative Route

Exercise 3.B.1: Counting Parameters

A fraud classifier (two classes) uses four attributes: **payment** (card, invoice, wallet), **country** (domestic, EU, other), **new device** (yes/no) and **night** (yes/no).

1. How many free numbers does the full generative model need — the class prior and one joint $p(x \mid y)$ per class?
2. How many does naive Bayes need?
3. For p yes/no attributes and K classes, give both counts as formulas.
4. From which p on does the full joint need more than a million numbers *per class*?

Exercise 3.B.1: Counting Parameters — Solution

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1. Full model. Each class needs a table over $3 \cdot 3 \cdot 2 \cdot 2 = 36$ value combinations, of which 35 are free (they sum to 1). Two classes give 70; the prior $P(\text{fraud})$ adds one: **71**.

2. Naive Bayes. Per class one categorical per attribute: $(3-1)+(3-1)+(2-1)+(2-1) = 6$ free numbers. Two classes give 12, plus the prior: **13**.

3. Formulas. Full: $K(2^p - 1) + (K - 1)$. Naive Bayes: $Kp + (K - 1)$.

4. $2^{19} - 1 = 524{,}287$ and $2^{20} - 1 = 1{,}048{,}575$, so from $p = 20$ on. The naive count grows linearly: 20 per class.

Exercise 3.B.2: Fitting Is Counting

An attribute takes k values. Within one class we observed it n times, value v exactly n_v times ($\sum_v n_v = n$). Model it as categorical with parameters $\theta_1, \ldots, \theta_k \geq 0, \sum_v \theta_v = 1$.

1. Write down the log-likelihood $\ell(\theta)$ of the n i.i.d. observations.
2. For $k = 2$, substitute $\theta_2 = 1 - \theta_1$ and show that the MLE is $\hat{\theta}_1 = n_1/n$ — the h/n of Lecture 03.
3. For general k, maximise ℓ under the constraint $\sum_v \theta_v = 1$ (Lagrange multiplier) and show $\hat{\theta}_v = n_v/n$.
4. Why can naive Bayes fit every attribute's table separately, while a full joint $p(x \mid y)$ could not be fitted attribute by attribute?

Exercise 3.B.2: Fitting Is Counting — Solution

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1. Each observation of value v contributes θ_v:

$$\ell(\theta) = \sum_{v=1}^{k} n_v \log \theta_v$$

2. $\ell = n_1 \log \theta_1 + n_2 \log(1-\theta_1)$. Setting $\ell'(\theta_1) = n_1/\theta_1 - n_2/(1-\theta_1) = 0$ gives $n_1(1-\theta_1) = n_2\theta_1$, so $\hat{\theta}_1 = n_1/(n_1+n_2) = n_1/n$.

3. $L(\theta, \lambda) = \sum_v n_v \log \theta_v - \lambda(\sum_v \theta_v - 1)$. From $\partial L/\partial\theta_v = n_v/\theta_v - \lambda = 0$ we get $\theta_v = n_v/\lambda$; the constraint gives $\lambda = \sum_v n_v = n$, so $\hat{\theta}_v = n_v/n$ — the relative frequency.

4. Under the naive assumption the log-likelihood of the table is a **sum** of one term $\sum_v n(x_j = v, y) \log \theta_{jvy}$ per attribute and class; no parameter appears in two terms, so each term is maximised on its own — by counting. A full joint has one parameter per *combination* of values; its counts are counts of whole rows, and they cannot be split into per-attribute counts.

3.C Naive Bayes by Hand

Exercise 3.C.1: Twelve Transactions

Flags: new device D, foreign IP F, night N, high amount H (1 = yes).

#	D	F	N	H	fraud?	#	D	F	N	H	fraud?
1	1	1	1	1	yes	7	1	0	0	0	no
2	1	1	0	1	yes	8	0	0	0	0	no
3	1	0	1	0	yes	9	0	1	0	0	no
4	0	1	1	1	yes	10	0	0	0	0	no
5	0	0	0	0	no	11	1	0	1	0	no
6	0	0	1	0	no	12	0	0	0	0	no

1. Estimate $P(\text{fraud})$ and $P(\text{flag} = 1 \mid \text{class})$ for all four flags by counting.
2. Compute $P(\text{fraud} \mid x)$ for $x_1 = (D, F, N, H) = (1, 0, 0, 0)$.
3. Do the same for $x_2 = (0, 0, 0, 1)$. What goes wrong, and why?
4. Repeat 2 and 3 with Laplace smoothing on the flags (keep the prior unsmoothed). Which result changes a lot, which only a little?

Exercise 3.C.1: Twelve Transactions — Solution

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1. $P(\text{fraud}) = 4/12 = 1/3$. Among the 4 frauds, each flag occurs 3 times: $3/4$ for D, F, N, H. Among the 8 honest: $D\ 2/8, F\ 1/8, N\ 2/8, H\ 0/8$.

2. Fraud: $\frac{1}{3} \cdot \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{256} \approx 0.0039$. Honest: $\frac{2}{3} \cdot \frac{2}{8} \cdot \frac{7}{8} \cdot \frac{6}{8} \cdot 1 = 0.1094$. $P(\text{fraud} \mid x_1) = 0.0039/(0.0039 + 0.1094) = 1/29 \approx 0.034$.

3. Fraud: $\frac{1}{3} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{1}{256}$. Honest: $\ldots \cdot P(H{=}1 \mid \text{honest}) = 0$. So $P(\text{fraud} \mid x_2) = 1$: a transaction without any red flag except a high amount is **certain** fraud, because no honest high amount occurred in eight rows. One zero count vetoes the other three flags, which all speak for honest.

4. Smoothed, $(\text{count} + 1)/(n + 2)$: fraud $4/6 = 2/3$ for every flag; honest $D\ 0.3, F\ 0.2, N\ 0.3, H\ 0.1$. x_1: fraud $\frac{1}{3} \cdot \frac{2}{3} \cdot (\frac{1}{3})^3 = 0.0082$, honest $\frac{2}{3} \cdot 0.3 \cdot 0.8 \cdot 0.7 \cdot 0.9 = 0.1008$, so $P = 0.075$ (from 0.034). x_2: fraud 0.0082, honest $\frac{2}{3} \cdot 0.7 \cdot 0.8 \cdot 0.7 \cdot 0.1 = 0.0261$, so $P \approx 0.24$ (from 1). With 4 and 8 rows the prior of one pseudo-count per value is not small, so even x_1 moves — smoothing matters most where the counts are smallest.

Exercise 3.C.2: Smoothing Is a Prior

A yes/no flag was seen h times among n frauds.

1. With a $\mathrm{Beta}(1,1)$ prior on $\theta = P(\mathrm{flag} \mid \mathrm{fraud})$, what is the posterior (Lecture 03)? Show that its **mean** is Laplace's $(h+1)/(n+2)$.
2. What is the posterior's **peak** for $h = 0$? What does that say about "keep the peak"?
3. "Add-α" smoothing uses $(h+\alpha)/(n+2\alpha)$. Which prior belongs to it? Compute the estimate for $h = 0, n = 100$ and $\alpha = 0.5, 1, 10$.
4. A merchant category with $k = 10$ values; one category never occurred in 100 frauds. What does Laplace smoothing give, and which prior is behind it?

Exercise 3.C.2: Smoothing Is a Prior — Solution

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1. Adding counts: $\mathrm{Beta}(1+h,\ 1+n-h)$, with mean $\frac{1+h}{(1+h)+(1+n-h)} = \frac{h+1}{n+2}$.

2. The mode of $\mathrm{Beta}(a,b)$ is $\frac{a-1}{a+b-2}$; here $\frac{h}{n} = 0$. Under a flat prior the peak *is* the MLE, so keeping the peak keeps the zero. Laplace smoothing silently switches to the mean.

3. $\mathrm{Beta}(\alpha,\alpha)$: posterior $\mathrm{Beta}(\alpha+h, \alpha+n-h)$, mean $(h+\alpha)/(n+2\alpha)$. For $h=0, n=100$: $\alpha = 0.5$: $0.5/101 \approx 0.0050$; $\alpha = 1$: $1/102 \approx 0.0098$; $\alpha = 10$: $10/120 \approx 0.083$. A larger α is a stronger belief that θ is near $1/2$ — worth 2α imagined observations.

4. $(0+1)/(100+10) = 1/110 \approx 0.0091$ — the posterior mean under a flat **Dirichlet** $(1,\ldots,1)$, the Beta for more than two values.